



SAMRAT POLY RESINS

Manufacturer of Coating & Composite Resins · Doraha, Ludhiana, Punjab, India

SAFETY DATA SHEET

Doc. Ref: SPR-SDS-UVR · Rev. 01 · Issue Date: 8 August 2026

UV RESIN

UV/LED-Curable Epoxy Acrylate · Crystal-Clear, Low Viscosity · Coatings, Adhesives & Craft

SINGLE-PART UV-CURE RESIN — NOT TWO-PART, NOT A POLYESTER RESIN

This product is a SINGLE-COMPONENT, UV/LED-curable epoxy acrylate resin — it is NOT a two-part epoxy system (no separate hardener, no mixing ratio), and it is NOT an unsaturated polyester/styrene resin. It cures by exposure to UV/LED light, not by mixing with MEKP, cobalt accelerator, or an amine-type hardener. None of the two-part mixing/exotherm guidance used for this company's Clear Casting Resin or Epoxy Art Resin, and none of the styrene/flammable-liquid data used for this company's polyester resin products, applies to this product.

UV Resin is an epoxy acrylate resin, per the manufacturer's TDS, that cures under UV/LED light exposure (365–405 nm, 30–60 seconds under lamp) to a crystal-clear, high-gloss finish. The manufacturer's TDS confirms low viscosity, self-levelling behaviour, excellent scratch resistance, low shrinkage, high surface hardness, and strong adhesion across multiple surfaces. Recommended applications include jewellery making and resin art, encapsulation, doming and electronics, adhesive/label/metal/wood coating, and crafts, coatings, and miniature models. It is supplied as a single, ready-to-cure product — no separate hardener or mixing step is described in the manufacturer's TDS.

No food-safety, FDA, food-contact, medical-device, formal chemical-resistance certification, or specific outdoor/weathering lifespan claim is made beyond what the manufacturer's TDS explicitly states.

SECTION 1: IDENTIFICATION

1.1 Product Identifier

Product Name	UV Resin
Product Type	Single-component, UV/LED-curable epoxy acrylate resin
Chemical Family	Epoxy acrylate (per manufacturer TDS) — not an unsaturated polyester/styrene resin, and not the same product as this company's two-part epoxy casting products
Product Code / Internal Reference	UVR
SDS Document Reference	SPR-SDS-UVR, Rev. 01

1.2 Relevant Identified Uses and Uses Advised Against

- Recommended use: Jewellery making and resin art, encapsulation, doming and electronics, adhesive/label/metal/wood coating, and crafts, coatings, and miniature models, cured under UV/LED light per the manufacturer's specification (365–405 nm).
- Uses advised against: Food-contact, medical, pharmaceutical, or other applications not specifically approved by the manufacturer; curing under light sources/wavelengths not consistent with the manufacturer's specification; any use not confirmed with Samrat Poly Resins technical staff.

1.3 Details of the Supplier of the Safety Data Sheet

Company Name	Samrat Poly Resins
--------------	--------------------

Registered Activity	Manufacturer of Coating & Composite Resins (ISO 9001:2015 certified)
Address	Oswal Road, Near GT Road, Doraha, Ludhiana — 141421, Punjab, India
GST No.	03ALBPS3446G1ZB
Website	samratpolyresins.in
Email	info@samratpolyresins.in

1.4 Emergency Telephone Number

Company Emergency Contact	+91 74638 81652
National Poison Information Centre (India)	1800-116-117 (Toll-Free, 24x7) — National Poisons Information Centre, AIIMS New Delhi

SECTION 2: HAZARD IDENTIFICATION

CLASSIFICATION NOT ESTABLISHED — EXACT FORMULATION NOT DISCLOSED

A specific GHS classification is NOT established for this product. The manufacturer's TDS confirms the general chemistry ('epoxy acrylate') but does not disclose the exact oligomer/monomer identity, any reactive diluent, or the photoinitiator package. Acrylate-based resins, as a well-documented industry-wide chemistry class, commonly cause skin and eye irritation and are recognised as capable of causing skin sensitisation — this is useful general context, not a confirmed classification for this specific product's exact formulation.

This product's described cure behaviour (direct UV/LED curing with no separate hardener or mixing step) is only chemically possible if a photoinitiator is compounded into the resin as supplied — free-radical UV curing does not occur without one. This is a reasoned technical inference from the product's described behaviour, not a TDS-confirmed ingredient disclosure. Photoinitiator packages commonly used in UV-curable resins are, as a class, recognised in the industry to sometimes carry their own irritant or sensitising hazards distinct from the base oligomer — again, general context, not a confirmed hazard for this specific, undisclosed photoinitiator.

2.1 Classification of the Substance or Mixture (GHS)

Not established. Jaspal (company chemist) should confirm the exact epoxy acrylate oligomer, any reactive diluent, and the photoinitiator identity/concentration so that a defensible classification can be assigned.

2.2 General Precautions (Pending Composition-Specific Classification)

- Avoid skin and eye contact — treat as a likely irritant and possible sensitiser pending confirmation, consistent with the general acrylate chemistry class.
- Avoid repeated or prolonged skin contact — acrylate chemistries, as a class, are well documented to cause allergic skin sensitisation with repeated exposure over time, even where single exposures cause no visible reaction.
- Use in an adequately ventilated area.
- Do not expose the uncured resin to unprotected skin or eyes for extended periods during handling, application, or the curing process.
- Curing under a UV/LED lamp introduces UV light exposure as a separate, equipment-related hazard, distinct from the resin's own chemical hazards — see Sections 7 and 8.
- Wear the PPE specified in Section 8 when handling this product.

SECTION 3: COMPOSITION / INFORMATION ON INGREDIENTS

Component	CAS No.	Function
Epoxy acrylate oligomer/resin (per manufacturer TDS)	Proprietary / not disclosed	Principal resin component and the base UV-curable oligomer. May include a reactive diluent monomer to achieve the stated low viscosity (450 cPs) — not confirmed as present or absent, and not individually identified in the manufacturer's TDS.
Photoinitiator package (inferred, not TDS-	Proprietary / not disclosed / not	Reasonably inferred to be present, based on

Component	CAS No.	Function
confirmed as a named ingredient)	confirmed	the product's described single-component UV/LED cure behaviour (365–405 nm, 30–60 seconds) with no separate hardener or mixing step disclosed — free-radical UV curing of an acrylate system requires a photoinitiator. This is a technical inference from described product behaviour, not a TDS ingredient disclosure. Identity, chemistry, and concentration are not established.

COMPOSITION NOT DISCLOSED — CONFIRM WITH JASPAL, INCLUDING PHOTOINITIATOR IDENTITY

Neither the exact epoxy acrylate oligomer identity nor the photoinitiator package is disclosed in the manufacturer's TDS. The photoinitiator's presence is inferred from the product's described behaviour, not confirmed as a named ingredient — its actual identity should be confirmed with Jaspal/the manufacturer, as photoinitiator chemistry can materially affect the product's hazard classification, PPE requirements, and UV-lamp compatibility.

SECTION 4: FIRST-AID MEASURES

4.1 Description of First-Aid Measures

Eye Contact	Rinse cautiously with clean water for several minutes. Remove contact lenses if present and easy to do — continue rinsing. Seek medical attention if irritation persists.
Skin Contact	Remove contaminated clothing promptly. Wash skin thoroughly with soap and water. Avoid prolonged or repeated skin exposure — acrylate-based resins, as a class, can cause allergic skin sensitisation with repeated contact. Seek medical advice for persistent irritation or rash.
Inhalation	Move to fresh air where vapours or mist have been inhaled (e.g. during spray application). Obtain medical attention for persistent or significant symptoms.
Ingestion	Rinse mouth. Do not induce vomiting unless directed by medical personnel. Obtain medical advice for significant ingestion or if symptoms develop.

4.2 Most Important Symptoms and Effects, Acute and Delayed

Not fully established given undisclosed exact composition. General acrylate-class precaution: skin and eye irritation is possible from direct contact; repeated skin exposure is a recognised route to allergic sensitisation for this chemistry class generally, though this has not been confirmed or ruled out for this specific product's exact formulation.

4.3 Notes to Physician

Treat symptomatically. No specific antidote is known or indicated by available product information. If sensitisation is suspected from repeated occupational exposure, evaluate accordingly.

SECTION 5: FIRE-FIGHTING MEASURES

Fire behaviour depends on the actual, undisclosed resin and photoinitiator composition. This product is not treated as a flammable-liquid/styrene-solution fire hazard by default — that classification belongs to this company's polyester resin products and does not automatically apply here. No flash point is disclosed, so flammability is treated as genuinely undetermined rather than assumed either way.

5.1 Extinguishing Media

- Suitable: Foam, dry chemical, carbon dioxide (CO₂), water fog/spray for cooling containers — select based on the surrounding fire and actual composition.

5.2 Special Hazards Arising from the Substance or Mixture

Combustion or decomposition of the organic resin and photoinitiator components may produce carbon monoxide, carbon dioxide, smoke, and irritating decomposition products. Styrene vapours and organic peroxide decomposition — relevant to this company's polyester resin and MEKP products — are NOT expected for this epoxy acrylate system and are not mentioned here as active hazards.

5.3 Advice for Fire-Fighters

- Use suitable protective equipment and self-contained breathing apparatus (SCBA) as a general precaution.
- Cool fire-exposed containers with water spray from a safe distance where practical.

SECTION 6: ACCIDENTAL RELEASE MEASURES

6.1 Personal Precautions, Protective Equipment and Emergency Procedures

- Restrict unnecessary access to the spill area. Provide adequate ventilation.
- Avoid skin and eye contact. Wear suitable PPE as described in Section 8 before attempting clean-up.
- Keep the spill shielded from direct sunlight/UV light during clean-up — significant UV exposure may begin curing the spilled material, making it harder to remove.

6.2 Environmental Precautions

- Prevent entry into drains and waterways. Contain the spill where safe to do so.
- Notify the relevant Pollution Control Board and site EHS personnel for any spill that reaches, or is likely to reach, a drain or waterway.

6.3 Methods and Material for Containment and Cleaning Up

- Absorb using a suitable inert absorbent. Transfer into correctly labelled waste containers.
- Clean promptly — this product is designed to cure under UV/LED light, and prolonged exposure of a spill to ambient UV (including strong daylight) may cause partial curing, complicating clean-up.
- Dispose through authorised waste channels — see Section 13.

SECTION 7: HANDLING AND STORAGE

7.1 Precautions for Safe Handling

- Avoid skin and eye contact. Maintain adequate ventilation, particularly during application or spray use.
- Keep containers closed when not in use, and shielded from direct sunlight/UV light during handling — this product can begin to cure on exposure to UV light, not only under the specified 365–405 nm lamp but potentially from strong ambient UV sources.
- If curing under a UV/LED lamp, follow the equipment manufacturer's guidance for lamp use — direct skin or eye exposure to UV/LED curing light is an equipment-related hazard distinct from the resin's own chemical hazards (see Section 8).
- Avoid prolonged or repeated skin contact. Maintain good workshop/industrial hygiene.

7.2 Conditions for Safe Storage, Including Any Incompatibilities

UV-SPECIFIC STORAGE REQUIREMENT — NOT JUST A HEAT PRECAUTION

This product must be stored protected from direct sunlight AND UV exposure specifically — not just general heat, as with this company's other resin products. Because this resin cures on UV light exposure by design, storage in unshielded daylight or near UV sources can cause the product to begin curing (thickening, gelling, or hardening) inside its own container, independent of any temperature effect. Opaque or UV-blocking packaging and a shielded storage area are recommended.

Storage Temperature	15–25°C, per manufacturer TDS.
Storage Conditions	Store in a cool, dry place, protected from direct sunlight and UV exposure, per manufacturer TDS. This is a stricter requirement than simple 'avoid sunlight' guidance — see callout above.

Shelf Life	12 months in unopened container, per manufacturer TDS.
Incompatible Materials	Not established given undisclosed composition. As a general precaution, keep away from strong oxidisers and avoid conditions (heat, UV/light exposure) that could trigger premature curing.
Packaging Material	200 kg pack, per manufacturer TDS — supplied at industrial bulk scale (MOQ 1,000 kg), distinct from the smaller-pack epoxy casting/art products in this company's range.

7.3 Specific End Use(s)

UV/LED-cured coatings, adhesives, jewellery/resin art, encapsulation, doming, and electronics applications. Refer to the product TDS (SPR-TDS-UVR) for the specified cure wavelength (365–405 nm) and cure time under lamp (30–60 seconds). This SDS does not provide detailed application or lamp-operating instructions.

SECTION 8: EXPOSURE CONTROLS / PERSONAL PROTECTION

No occupational exposure limit is established for this product, as its composition (including the inferred photoinitiator) is not disclosed.

8.2 Exposure Controls

- Engineering controls: General ventilation; local exhaust ventilation where vapour or mist may occur (e.g. spray application).
- Eye/face protection: Chemical splash goggles for handling the liquid resin. Separately, if working near an operating UV/LED curing lamp, UV-protective eyewear appropriate to the lamp's actual wavelength output should be used per the lamp manufacturer's guidance — this is an equipment-safety measure, not a resin-chemistry PPE requirement, and is outside the scope of what this SDS can specify without lamp-specific data.
- Hand protection: Chemical-resistant gloves suited to acrylate-resin handling (e.g. nitrile is a common general-purpose choice for acrylate chemistry, though glove selection should ideally be confirmed against this product's actual composition once known).
- Body protection: Protective clothing, changed promptly if contaminated. Given the sensitisation potential common to acrylate chemistry, minimise bare-skin contact rather than relying on PPE alone.
- Respiratory protection: Use appropriate respiratory protection where ventilation is inadequate, particularly for spray application.
- Skin covering during UV/LED lamp curing: direct, unprotected skin exposure to UV/LED curing light sources is a separate, equipment-related hazard (comparable in principle to other UV light sources) — follow the lamp manufacturer's operating guidance for skin protection during the curing step.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

Appearance	Clear liquid; crystal-clear cured finish, per manufacturer TDS
Odour	Not established — no odour data provided in the manufacturer's TDS
pH	Not established — no product-specific pH data provided
Melting / Freezing Point	Not established
Initial Boiling Point / Range	Not established
Flash Point	Not established — no product-specific value provided in the manufacturer's TDS. Genuinely undetermined; NOT assumed to be flammable or non-flammable, and NOT assumed to match this company's polyester resin flash-point range
Evaporation Rate	Not established
Vapour Pressure	Not established
Vapour Density	Not established
Solubility in Water	Not established — no product-specific data provided

Auto-Ignition Temperature	Not established
Viscosity	450 cPs at 25°C, per manufacturer TDS — single-component product; no ambiguity between mixed/unmixed states applies here, unlike this company's two-part epoxy products
Curing Type	UV/LED cure, per manufacturer TDS — not a chemical hardener/mixing cure
Cure Wavelength / Energy	365–405 nm, per manufacturer TDS
Cure Time (Under Lamp)	30–60 seconds, per manufacturer TDS
Relative Density / Specific Gravity	1.10 g/cm ³ , per manufacturer TDS (chemist-verified)
Cured Hardness	Shore D 80, per manufacturer TDS — a cured-product performance characteristic, not a liquid property
Surface Properties (Cured)	Excellent scratch resistance, low shrinkage, per manufacturer TDS — cured-product performance characteristics
VOC Content	Not established — no product-specific VOC data provided; not invented

SECTION 10: STABILITY AND REACTIVITY

10.1 Reactivity

Stable under normal storage conditions when shielded from UV light and excessive heat, per available product information. This product is specifically designed to react (cure) upon exposure to UV/LED light at 365–405 nm — this is the intended function of the product, not a hazardous reaction in itself, but uncontrolled/premature exposure during storage or handling should be avoided (see Section 7 callout).

10.2 Chemical Stability

Stable under recommended storage conditions (cool, dry, shielded from UV light and direct sunlight). Unlike this company's polyester resin products (heat/contamination-driven instability) or two-part epoxy products (mixing-driven instability), this product's principal stability concern is UV/light exposure specifically.

10.3 Possibility of Hazardous Reactions

None expected under normal handling and storage conditions when shielded from UV light. The intended UV/LED curing reaction, when carried out as directed, is not considered a hazardous reaction — it is the product's designed function.

10.4 Conditions to Avoid

- Exposure to UV light or strong direct sunlight during storage or handling, outside the intended, controlled curing step.
- Excessive heat during storage.
- Contamination with incompatible materials (not established given undisclosed composition).

10.5 Incompatible Materials

Not established given undisclosed composition. The MEKP-contamination and cobalt-promoter incompatibility warnings used for this company's polyester resin products, and the Part A/Part B mixing-exotherm warnings used for this company's two-part epoxy products, do NOT apply here — this product uses neither of those cure mechanisms.

10.6 Hazardous Decomposition Products

Not established. Under fire or excessive heat, organic resin and photoinitiator components may generate carbon monoxide, carbon dioxide, smoke, and irritating decomposition products — see Section 5.

SECTION 11: TOXICOLOGICAL INFORMATION

11.1 Information on Toxicological Effects

Acute Toxicity — Oral/Inhalation/Dermal	Not established
Skin Corrosion / Irritation	Not established for this exact formulation. General class-level context: acrylate-based resins commonly cause skin irritation in the wider industry — not confirmed for this specific product
Serious Eye Damage / Irritation	Not established for this exact formulation. General class-level context: acrylate-based resins commonly cause eye irritation in the wider industry — not confirmed for this specific product
Respiratory / Skin Sensitisation	Not established for this exact formulation. General class-level context: acrylate chemistry is well-documented industry-wide as capable of causing skin sensitisation with repeated exposure — a reasonable general expectation for this chemistry class, not a confirmed classification for this specific product
Germ Cell Mutagenicity	Not established
Carcinogenicity	Not established — not asserted without composition support
Reproductive Toxicity	Not established — not asserted without composition support
STOT — Single/Repeated Exposure	Not established
Aspiration Hazard	Not asserted — composition supporting this classification is not confirmed

SECTION 12: ECOLOGICAL INFORMATION

Not established for the complete product. Styrene aquatic toxicity data and the H412 classification used for this company's polyester resin products do NOT apply here.

12.1 General Precaution

Prevent release into drains, soil, groundwater, and surface water, consistent with good practice for any industrial/craft chemical product regardless of formal ecological classification.

12.2 Persistence, Bioaccumulation, and Other Effects

Not established. A formal PBT/vPvB assessment has not been performed.

SECTION 13: DISPOSAL CONSIDERATIONS

13.1 Waste Treatment Methods

- Dispose of unused resin through authorised industrial waste channels.
- Do not discharge uncured resin to drains or waterways.
- Empty containers may retain resin residue. Handle contaminated packaging appropriately.
- Fully UV-cured waste material is generally an inert solid; confirm current requirements with the relevant State Pollution Control Board for the specific cured composition.
- Follow all applicable local, state, and national regulations for disposal.

SECTION 14: TRANSPORT INFORMATION

TRANSPORT — NOT ESTABLISHED, NOT ASSUMED SAFE

Transport classification is NOT established. This is genuinely undetermined, not assumed to be 'not regulated.' Do NOT use UN 1866 / RESIN SOLUTION / Class 3 / Packing Group III — that classification belongs to this company's polyester resin products and does not apply here. Do NOT use UN 3105 / Organic Peroxide Type D / Class 5.2 — that belongs to

MEKP Hardener, a separate and unrelated product.

UN Number	Not established — requires actual composition data, including the photoinitiator identity
Special Precautions for User	Transport in tightly closed, properly labelled, UV-shielded/opaque packaging to prevent both leakage and premature light-triggered curing.

SECTION 15: REGULATORY INFORMATION

15.1 India

- General workplace chemical-safety requirements under the Factories Act, 1948 and applicable State Factory Rules apply to industrial handling of this product, as they would to any industrial chemical mixture.
- This product is not subject by default to the styrene-specific occupational exposure requirements, flammable-liquid handling/storage rules under MSIHC 1989, or organic peroxide provisions applied to this company's polyester resin and MEKP products — those apply to chemically different products. Once the exact resin and photoinitiator composition is confirmed, this section should be revisited.
- If UV/LED lamps are used for curing, applicable electrical/equipment safety and workplace UV-exposure requirements for the lamp itself are a separate consideration from this resin's chemical regulatory status.
- Users must comply with all applicable local, state, national, and transport regulations for the actual composition of this product.

15.2 Claims Not Made

This SDS does not claim food safety, FDA approval, food-contact approval, medical-device compliance, formal chemical-resistance certification, or a specific outdoor/weathering lifespan. Any such claim requires actual manufacturer test/specification documentation, which is not included in the TDS reviewed for this SDS.

SECTION 16: OTHER INFORMATION

Abbreviations

CAS = Chemical Abstracts Service · GHS = Globally Harmonized System of Classification and Labelling of Chemicals · LED = Light-Emitting Diode · MEKP = Methyl Ethyl Ketone Peroxide (a different, unrelated product) · nm = nanometre · PBT/vPvB = Persistent, Bioaccumulative and Toxic / very Persistent, very Bioaccumulative · SDS = Safety Data Sheet · TDS = Technical Data Sheet · UV = Ultraviolet.

Revision Information

SDS Version / Revision No.	01
Issue Date	8 August 2026
Prepared By	Samrat Poly Resins / Regulatory Department
Reason for Issue	Initial official issue

This SDS has been prepared based on available technical data, manufacturer/chemist-reviewed product information, component hazard data, and applicable safety references. The information is provided for safe handling, storage, transport, use, and disposal of the product. Several sections cannot be completed beyond 'not established' because the manufacturer's TDS discloses the general chemistry family (epoxy acrylate) and performance specifications but does not disclose the exact oligomer, reactive diluent (if any), or photoinitiator composition. Users are responsible for confirming suitability for their specific application and for compliance with current regulations at the time of use.